BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type (Group-I)

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, Lahore, 2021. Session 2017-2019 to 2020-2022. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Density of an ideal gas can be calculated by the formula:

(A) d = nRT
(B) d = PM/RT
(C) d = m/M RT
(D) d = PV/M

Q2. Which of the following species has unpaired electrons in anti-bonding molecular orbitals:

(A) O_2^{2+}
(B) N_2^{2-}
(C) B_2
(D) F_2

Q3. The pH of mixture of CH_3COONa and CH_3COOH is:

(A) 7
(B) >7
(C) <7
(D) 1

Q4. The unit of rate constant is the same as that of rate of reaction in:

(A) First order reaction
(B) Second order reaction
(C) Zero order reaction
(D) Third order reaction

Q5. Many elements have fractional atomic masses, this is because:

(A) The mass of an atom is itself fractional

(B) Atomic masses are average masses of isobars

(C) Atomic masses are average masses of isotopes

(D) Atomic masses are average masses of isotopes proportional to their relative abundance

Q6. In the monoclinic crystal system, bond axes are:

(A) $a = b = c$

(B) $a = b \neq c$

(C) $a \neq b = c$

(D) $a \neq b \neq c$

Q7. Which of the following compounds do not show process of sublimation:

(A) Ammonium chloride

(B) Iodine

(C) Naphthalene

(D) Carbon tetra chloride

Q8. In endothermic reactions, the heat content of the:

(A) Products is more than that of reactants

(B) Reactants is more than that of products

(C) Surrounding increases

(D) Reactants and products are equal

Q9. If a salt bridge is not used between two half cells, then the voltage:

(A) Decreases rapidly

(B) Decreases slowly

(C) Does not change

(D) Drops to zero

Q10. Orbitals having same energy are called:

(A) Hybrid orbitals

(B) Valence orbitals

(C) Degenerate orbitals

(D) d-orbitals

Q11. 1 gram formula of NaCl, is equal to:

(A) 58.5 g

(B) 23 g

(C) 35.5 g

(D) 12 g

Q12. Balmer series in hydrogen spectrum lies in the region:

(A) Ultraviolet

(B) Visible

(C) Infrared

(D) Microwave

Q13. When water freezes at 0°C, its density decreases due to:

(A) Cubic structure of ice

(B) Empty spaces present in structure of ice

(C) Change of bond lengths

(D) Change of bond angles

Q14. According to VSEPR theory, the shape of PFR molecule is:

(A) Trigonal pyramidal

(B) Tetragonal

(C) Linear

(D) Trigonal planer

Q15. If absolute temperature of a gas is doubled and the pressure is reduced to one half, the volume of the gas will:

(A) Remain unchanged
(B) Increase four times
(C) Reduce to ¼
(D) Be doubled

Q16. An excess of silver nitrate in aqueous form is added to aqueous barium chloride and precipitate is removed by filtration. What are main ions in the filtrate:

(A) Ag^{+} and NO_3^{-} only
(B) Ag^{+} and Ba^{2+} and O_3^{-}
(C) Ba^{2+} and NO_3^{-} only
(D) Ba^{2+} and O_3^{-} and Cl^{-}

Q17. A thermometer used in Landsberger's method can read upto:

(A) 0.1 K
(B) 0.01 F
(C) 0.01 K
(D) 0.01°C

Essay Type (Group-I)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: Session 2017-2019 to 2020-2022

QSECTION - I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i How is the law of conservation of mass obeyed during stoichiometric calculations?

ii Why N_2 and CO have same number of electrons, protons and neutrons? Justify.

iii Define mole. Calculate the gram atoms (moles) in 0.1 g

Note: Items iv–viii of Q.2 are not visible in the source scan.

ix State Charles's law. Write down its mathematical form.

x Relative lowering of vapour pressure is independent of temperature. Justify this statement.

xi Define hydration energy of ions.

xii What are continuous solubility curves? Give one example.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i What is role of hydrogen bonding in paints, dyes, and textile materials?

ii What do you mean by liquid crystal? Write down any two uses of it.

iii Define the property of solids allotropy and give two examples.

iv The crystals showing isomorphism mostly have the same atomic ratios; explain.

v How neutron was discovered by Chadwick? Also write down reaction.

vi Write down postulates of Bohr's atomic model.

vii How azimuthal quantum number (l) gives information about types of subshells?

viii Explain the concept of atomic spectrum.

ix Write down optimum conditions of temperature and pressure in the manufacture of ammonia by Haber's process.

x Define pH and pOH of solutions.

xi What do you understand by rate determining step? Give a suitable example.

xii How does Arrhenius equation help us to calculate the energy of activation of a reaction?

QQ.4. (Marks 12) Write short answers to any SIX (6) questions.

i Define ionization potential of element. How ionization potential vary across the period?

ii Anionic radius is greater than that of its parent atomic radius. Why?

iii Draw the structure of NH_3 with reference to VSEPR Theory.

iv How do electronegativity values charge in a group?

v Define enthalpy of solution with an example.

vi State first law of thermodynamics. Give its mathematical expression.

vii Calculate the oxidation numbers of elements underlined: (a) Na_2CO_3 (b) K_2MnO_4

viii Give function of salt bridge.

ix Why SHE acts as cathode when connected with Zn electrode but SHE acts as anode when connected with Cu? Justify.

QSECTION-II. Note: Attempt any three questions.

Q5(a). (Marks 4) When limestone ($CaCO_3$) is roasted then quicilime (CaO) is formed according to the following equation. The actual yield of (CaO) is 2.5kg, when 4.5 kg of limestone is heated. What is the percentage yield of this reaction? $CaCO_3(s) \rightarrow \Delta CaO(s) + CO_2(g)$

Q5(b). (Marks 4) Discuss the role of Hydrogen Bonding in Biological Compounds.

Q6(a). (Marks 4) Write fundamental postulates of kinetic molecular theory of gases.

Q6(b). (Marks 4) Discuss four postulates of Bohr's model of atom.

Q7(a). (Marks 4) What is Sp^3 hybridization? Explain the structure of methane.

Q7(b). (Marks 4) Explain measurement of enthalpy by a glass calorimeter.

Q8(a). (Marks 4) Calculate the pH of 1.0 mole dm^{-3} of NH_4OH, which is 1% dissociated.

Q8(b). (Marks 4) Explain half life method for determination of order of reaction.

Q9(a). (Marks 4) Freezing points of solutions are depressed when non-volatile solutes are present in volatile solvents. Justify it. Plot a graph to elaborate your answer.

Q9(b). (Marks 4) Discuss measurement of electrode potential by standard hydrogen electrode (S.H.E)

GROUP – II

Objective Type (Paper-I, Group-II)

Time Allowed: 20 Min. | **Maximum Marks:** 17

> Note: CHEMISTRY, Inter-I, 2021, Paper-I, Group-II. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. An excess of aqueous silver nitrate is added to aqueous barium chloride and precipitate is removed by filtration. What are the main ions in the filterate?

(A) Ag^+ and NO_3^- only
(B) Ag^+ and Ba^{2+} and NO_3^-
(C) Ba^+ and NO_3^- only
(D) Ba^{2+} and NO_3^- and Cl^-

Q2. Pressure remaining constant, at which temperature the volume of a gas will become twice of what it is at 0°C.

(A) 546°C
(B) 200°C
(C) 546K
(D) 273K

Q3. The number of bonds in nitrogen molecule is:

(A) one σ and one π
(B) one σ and two π
(C) three sigma only
(D) two σ and one π

Q4. When water freezes at 0°C, its density decreases due to:

(A) cubic structure of ice
(B) empty spaces present in the structure of ice
(C) change of bond lengths
(D) change of bond angles

Q5. Isotopes differ in:

(A) properties which depend upon mass
(B) arrangement of electrons in orbitals
(C) chemical properties
(D) the extent to which they may be affected in electromagnetic field

Q6. 18g glucose is dissolved in 90g of water. The relative lowering of vapour pressure is equal to:

(A) 1/5
(B) 5.1
(C) 1/51
(D) 6

Q7. The nature of the positive rays depend on:

(A) the nature of electrode
(B) the nature of the discharge tube
(C) the nature of the residual gas
(D) all of these

Q8. The velocity of photon is:

(A) independent of its wavelength
(B) depends on its wavelength
(C) equal to square of its amplitude
(D) depends on its source

Q9. One calorie is equivalent to:

(A) 0.4184 J
(B) 41.84 J
(C) 4.184 J
(D) 418.4 J

Q10. Acetone and chloroform are soluble in each other due to:

(A) intermolecular hydrogen bonding
(B) ion-dipole interaction
(C) instantaneous dipole
(D) all of these

Q11. Solvent extraction is an equilibrium process and it is controlled by:

(A) law of mass action
(B) the amount of solvent used
(C) distribution law
(D) the amount of solute

Q12. In zero order reaction, the rate is independent of:

(A) temperature of reaction
(B) concentration of reactants
(C) concentration of products
(D) none of these

Q13. Which of the following species has unpaired electrons in antibonding molecular orbitals?

(A) O_2^{2+}
(B) N_2^{2-}
(C) B_2
(D) F_2

Q14. 27g of Al react completely with how much mass of O_2 to produce Al_2O_3.

(A) 8g of oxygen
(B) 16g of oxygen
(C) 24g of oxygen
(D) 32g of oxygen

Q15. The cathodic reaction in electrolysis of dil. H_2SO_4 with Pt electrodes is:

(A) reduction
(B) oxidation
(C) both oxidation and reduction
(D) neither oxidation nor reduction

Q16. The molar volume of CO_2 is maximum at:

(A) STP
(B) 127°C and 1 atm
(C) 0°C and 2 atm
(D) 273°C and 2 atm

Q17. The pH of 10^{-3} mol dm^{-3} of an aqueous solution of H_2SO_4 is:

(A) 3.0
(B) 2.7
(C) 2.0
(D) 1.5

Subjective Type (Paper-I, Group-II)

Time Allowed: 2:40 Hrs. | **Maximum Marks:** 68

QSECTION-I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT(8) questions.

i What is gram formula? Give example.

ii Define stoichiometry, write down its two laws.

iii How limiting reactant is identified?

iv What is distribution law?

v What is mobile phase and stationary phase?

vi What is physical meaning of value R?

vii What is Avogadro's law? Give example.

viii Where plasma is found?

ix How pressure of dry gas is calculated?

x Define solubility curve, give its types.

xi Give two differences between ideal and non-ideal solutions.

xii What is fractional crystallization?

QQ.3. (Marks 16) Write short answers to any EIGHT(8) questions.

i Explain cleavage plane is anisotropic property.

ii Amorphous solids like glass are also called super cooled liquids. Explain.

iii Define isomorphism by giving on example.

iv Explain why HF is weak acid than HI?

v Define Zeeman effect and stark effect.

vi State Heisenberg's uncertainty principle, write down its mathematical form.

vii What is spin quantum number? Give its significance.

viii What is difference between orbit and orbital?

ix Write down equilibrium constant expression for the reaction: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

x Calculate pH of 0.001 M HCl solution.

xi Explain the radioactive decay is 1st order reaction.

xii Justify the statement "the unit of rate constant of a second order reaction is dm^3 $mole^{-1}s^{-1}$ but the unit of rate of reaction is mole dm^{-3} s^{-1}"

QQ.4. (Marks 12) Write short answers to any SIX (6) questions.

i Name the four factors affecting ionization energies.

ii Why ionization energy decreases down the group inspite of the increase in proton number?

iii Why second ionization energy is higher than first ionization energy?

iv Define electron affinity with an example.

v Define first law of thermodynamics.

vi Define heat and work.

vii Calculate oxidation state of chromium in dichromate ion.

viii What is the use of salt bridge in voltaic cell?

ix Why voltaic cell is a reversible cell?

QSECTION-II. Note: Attempt any three questions.

Q5(a). (Marks 4) Calculate the gram atoms (moles) in: (i) 0.1g of sodium (ii) 0.1kg of silicon

Q5(b). (Marks 4) Explain the following properties of crystalline solids. Give two examples in each case: (i) Isomorphism (ii) Transition Temperature

Q6(a). (Marks 4) Derive an equation to find out the partial of a gas knowing the individual moles of component gases and the total pressure of the mixture.

Q6(b). (Marks 4) Give the postulates of Bohr's atomic model. Which postulate tells us that orbits are stationary and energy is quantized?

Q7(a). (Marks 4) Define electron affinity. Name the factors affecting on it. How does it vary in the periodic table.

Q7(b). (Marks 4) State first law of thermodynamics. Write down its mathematical expression. Prove that DΔ = q_p.

Q8(a). (Marks 4) What is the percentage ionization of acetic acid in a solution in which 0.1 mol of it has been dissolved per dm^3 of the solution. ($K_a = 1.85 \times 10^{-5}$)

Q8(b). (Marks 4) Explain half life method and large excess method to find the order of reaction.

Q9(a). (Marks 4) Freezing points of solutions are depressed when non-volatile solutes are present in volatile solvents. Justify it. Plot a graph to elaborate your answer.

Q9(b). (Marks 4) Write down the various rules for assigning oxidation number.